

Unit 2 Seminar: Threat Modelling Using STRIDE, Attack Trees, and MITRE ATT&CK

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A large international airport in the United States represents a complex cyber-physical system that relies on tightly integrated information technology (IT) and operational technology (OT) systems. These include passenger processing platforms, baggage handling systems, security infrastructure, and airline integration services. Due to the critical nature of airport operations, the environment requires strong guarantees of confidentiality, integrity, and especially availability.

From a threat modelling perspective, the airport's key assets include passenger personally identifiable information (such as passport and payment data), flight scheduling and operational systems, baggage handling infrastructure, and security systems such as CCTV and access control. Any disruption or compromise of these assets could result in significant operational, financial, and safety impacts.

Using the STRIDE methodology (Microsoft, 2009), several key threats can be identified. Spoofing threats include attackers gaining access through stolen staff credentials or impersonating third-party vendors. Tampering may involve manipulation of flight schedules or baggage routing systems. Repudiation risks arise where insufficient logging allows users to deny malicious actions. Information disclosure threats include leakage of passenger data or exposure of misconfigured systems. Denial of service attacks, such as ransomware or distributed denial-of-service (DDoS), could disrupt check-in or baggage operations. Elevation of privilege may occur if attackers exploit weak access controls or compromised accounts to gain administrative or operational system access.

An attack tree approach (Schneier, 1999) shows how an attacker's goal of disrupting airport operations could be achieved through multiple paths, including phishing staff credentials, exploiting vulnerabilities in web-facing systems, or targeting operational technology such as baggage handling

systems. External service disruption through API attacks or DDoS activity also represents a viable attack vector.

Relevant MITRE ATT&CK techniques (MITRE, 2024) include phishing (T1566), credential access via credential dumping (T1003), lateral movement through remote services (T1021), and impact through data encryption for disruption (T1486). These techniques reflect realistic adversary behaviours observed in real-world cyber incidents targeting critical infrastructure.

Key risks in this environment stem from the high level of interdependence between IT and OT systems, the reliance on third-party vendors, and the potential for cascading failures across operational services. Even minor disruptions can have significant downstream effects on airport safety and efficiency.

To mitigate these risks, several controls are recommended, including strong network segmentation between IT and OT environments, multi-factor authentication for all privileged accounts, strict vendor access controls using least privilege principles, continuous monitoring through security information and event management (SIEM) systems, and regular penetration testing. These measures help reduce the likelihood and impact of successful attacks.

Overall, the airport environment demonstrates the importance of layered threat modelling approaches. Combining STRIDE with attack trees and MITRE ATT&CK provides a structured and realistic understanding of threats in complex critical infrastructure systems. This aligns with guidance from OWASP (2024) and the Threat Modelling Manifesto principles, which emphasise iterative and collaborative security design.

References:

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